DSI Scholar Program Project Proposal

Research Track: Economic History & Data Science

General Information

Project Title:

From Pumps to Plugs: The Green Revolution and the Structural Transformation of Rural Electricity Demand

Public Project Summary

Does rural electrification lead to economic growth, or is it a byproduct of it? This project investigates the causal link between agricultural productivity and the structural transformation of rural energy markets. While traditional literature often views household electrification as the primary driver of development, our research explores a "productivity-first" pathway. We hypothesize that the **Green Revolution** provided the initial impetus for large-scale rural electrification through the demand for irrigation pumps. This "productive" electrification created a physical infrastructure that, with a significant time lag, lowered the marginal cost of connecting households and stimulated long-run consumer demand.

Using a historical lens (1960–2000), we examine whether high-value agricultural sectors—rather than residential lighting—are the true catalysts for permanent structural change. The study utilizes a **Shift-Share Instrumental Variable (IV)** design and advanced **OCR/HTR technologies** to digitize handwritten archives. By analyzing outcomes such as household assets, income levels, and shifts toward urban employment, we aim to clarify whether electricity is a driver of transformation or a follower of productivity.

DSI Scholar Responsibilities

- **Digitization & OCR Pipeline:** Implementing and refining OCR/HTR workflows (Tesseract, Google Cloud Vision) to extract tabular data from digitized, handwritten PDF archives of historical district census reports.
- **Data Harmonization:** Merging newly digitized historical records with existing electronic datasets, resolving district boundary changes over 40 years.
- **Econometric Support:** Assisting in the development of a shift-share instrument interacting crop-specific shocks with baseline district shares.

- **Visualization:** Creating spatial-temporal maps showing the diffusion of the electricity grid from agricultural clusters to residential zones.

Project Plan & Timeline

Phase	Tasks	Hours
Weeks 1-4	**Data Extraction & OCR:** Set up OCR/HTR pipeline for handwritten PDFs. Manual cleaning of high-error samples.	40
Weeks 5-7	**Data Harmonization:** Merging extracted data with existing datasets; resolving boundary changes.	30
Weeks 8-10	**Econometric Modeling:** Constructing the Shift-Share IV and developing the first-stage model.	30
Weeks 11-12	**Secondary Analysis:** Measuring long-run household outcomes (assets, employment) and testing the "Productivity vs. Price" hypothesis.	20
Weeks 13-14	**Documentation & Delivery:** Finalizing report, documenting OCR methodology, and preparing visualizations.	20
Total Estimated Hours		**140**

Requirements & Impact

Required Skills and Experience

- Proficiency in **Python** (OCR pipelines/CV) and **R or Stata** (econometrics).
- Experience with unstructured data or OCR libraries (Tesseract, OpenCV).
- Understanding of **Instrumental Variables (IV)** and causal inference.
- Strong attention to detail for data validation of historical records.

DSI Scholars Support Impact

> This project involves a significant "Data Engineering" challenge: the conversion of handwritten, non-machine-readable historical archives into a structured panel dataset. A DSI Scholar with skills in OCR and computer vision is uniquely positioned to unlock this data, transforming archived PDFs into a rigorous evidentiary base. This bridge between "archival data" and "modern econometric rigor" is essential for validating the long-term impact of productivity shocks on structural transformation.

Logistics

[X] Data already accessible

[X] International Students (F1/J1) eligible

[X] Student Levels: Advanced Undergraduate, Master's, PhD